IEEE - Institute of Electrical and Electronics Engineers

What is the IEEE?

- international non-profit, professional organization for the advancement of technology related to electricity.
- largest technical professional organization in the world (in number of members), with more than 360,000 members in around 175 countries (2005)

What does the IEEE do?

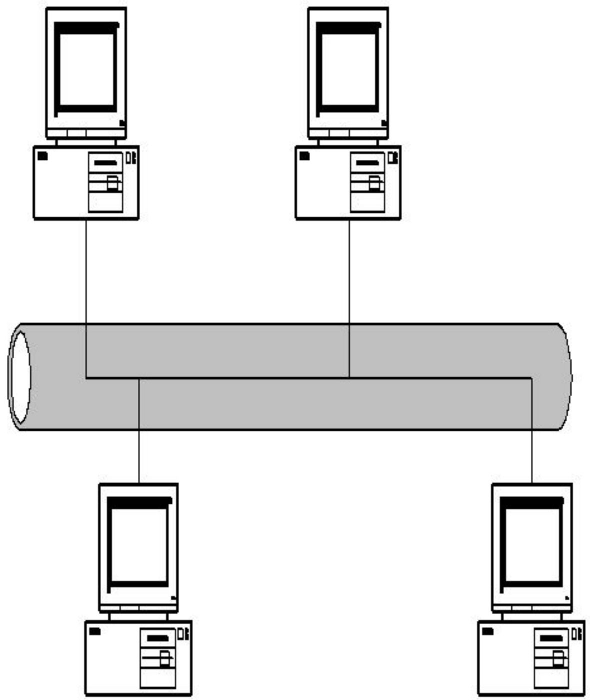
- produces 30 percent of the world's literature in the electrical and electronics engineering and computer science field,
- sponsors or cosponsors more than 300 international technical conferences each year.
- publishes an extensive range of peer-reviewed journals,
- major international standards body (nearly 900 active standards with 700 under development).

IEEE 802 Working Groups

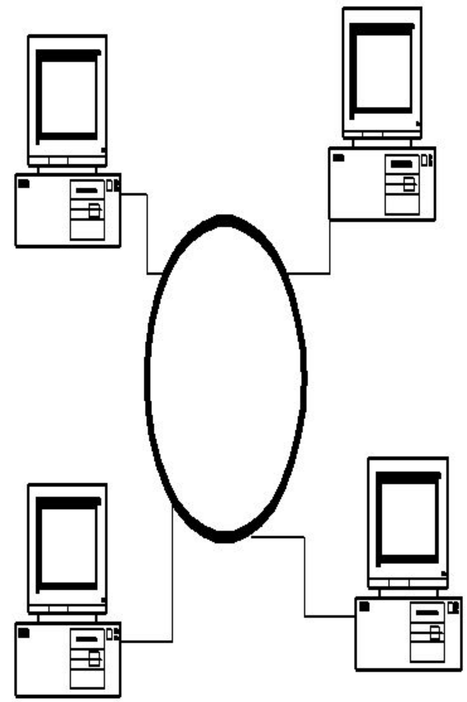
Active working groups	Inactive or disbanded working groups
802.1 Higher Layer LAN Protocols Working Group	802.2 Logical Link Control Working Group
802.3 Ethernet Working Group	802.4 Token Bus Working Group
802.11 Wireless LAN Working Group	802.5 Token Ring Working Group
802.15 Wireless Personal Area Network (WPAN) Working Group	802.7 Broadband Area Network Working Group
802.16 Broadband Wireless Access Working Group	802.8 Fiber Optic TAG
802.17 Resilient Packet Ring Working Group	802.9 Integrated Service LAN Working Group
802.18 Radio Regulatory TAG	802.10 Security Working Group
802.19 Coexistence TAG	802.12 Demand Priority Working Group
802.20 Mobile Broadband Wireless Access (MBWA) Working Group	802.14 Cable Modem Working Group
802.21 Media Independent Handoff Working Group	
802.22 Wireless Regional Area Networks	

Local Area Networks

- Local area networks (LANs) connect computers within a building or a enterprise network
- Almost all LANs are broadcast networks
- Typical topologies of LANs are **bus** or **ring** or **star**
- We will work with Ethernet LANs.



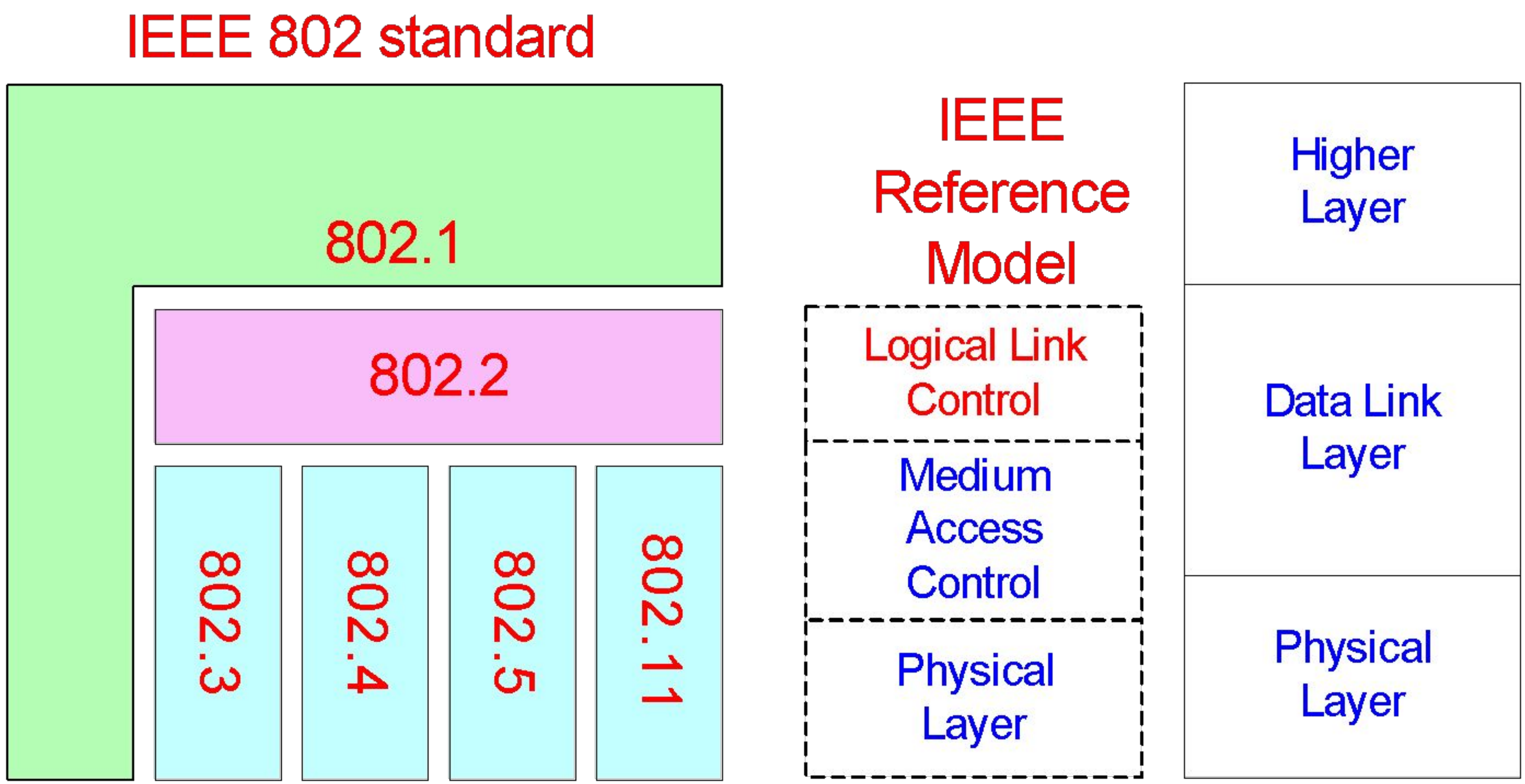
Bus LAN



Ring LAN

IEEE 802 Standards

- IEEE 802 is a family of standards for LANs, which defines an LLC and several MAC sub layers


IEEE 802 standard
802.1
802.2
802.3
802.4
802.5
802.11
IEEE
Reference
Model
Logical Link
Control
Medium
Access
Control
Physical
Layer
Higher
Layer
Data Link
Layer
Physical
Layer

IEEE 802 LAN Layers

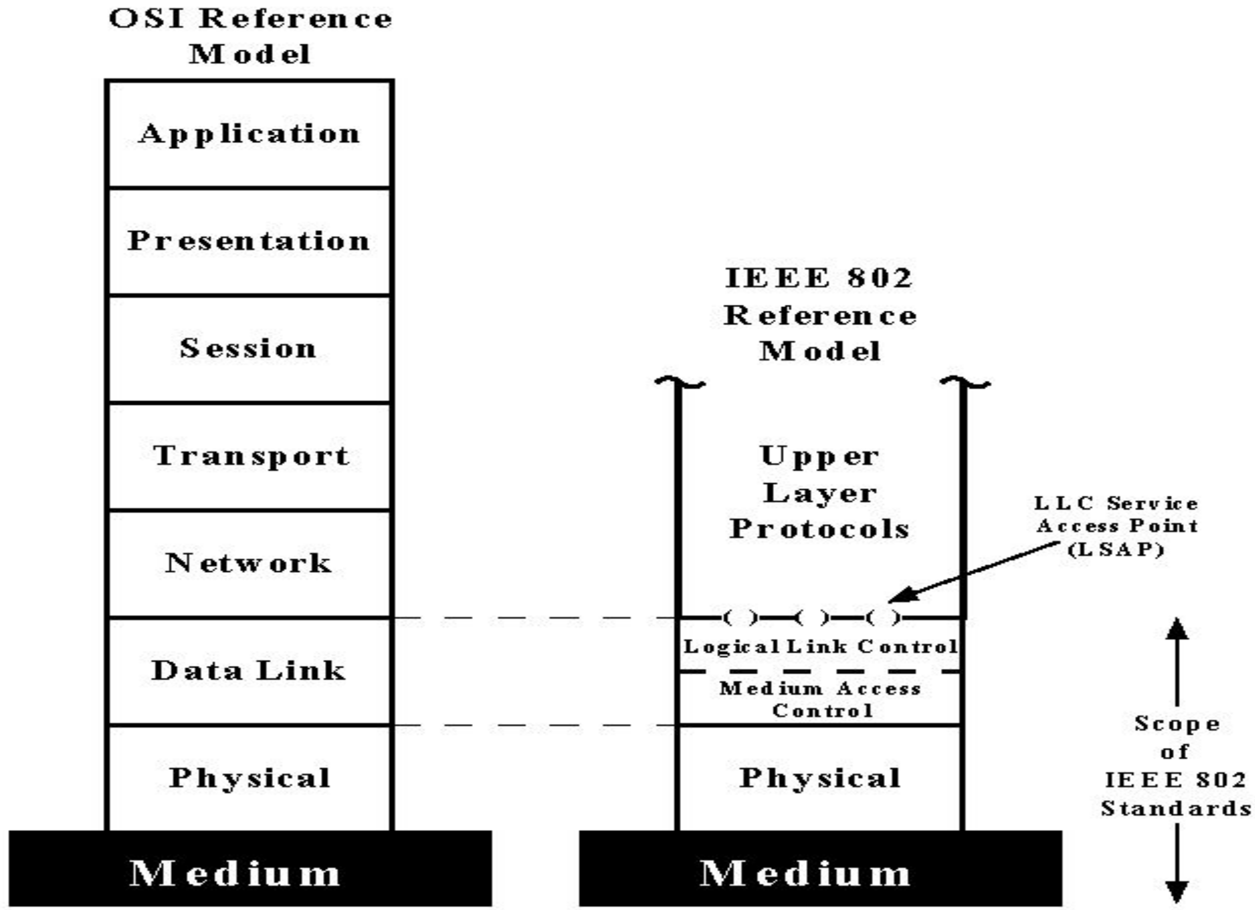


Figure 5.1 IEEE 802 Protocol Layers Compared to OSI Model

- IEEE 802.3 Ethernet LANs
  - The MAC layer uses CSMA/CD (Carrier Sense Multiple Access with Collision Detection) technology.
    - When a computer wants to transmit a frame it:
      - Listens for a frame on the cable, if busy, the computer waits for a random time and attempts transmission again. This is known as **Carrier Sense**.
      - If the cable is quiet, the computer begins to transmit.
      - Two computers could transmit at the same time. To prevent this happening, the transmitting computer listens to what it is sending.
      - If what it hears is different to what it is sending, then a collision has occurred. This is known as **Collision Detection**.

- Contd……..
  - When a computer wants to receive a frame it:
    - Listens to all frames traveling on the cable.
    - If the frame address is the same as the computer's address or the same as the group address of the computers of which it is a member, it copies the frame from the cable.
    - Otherwise it just ignores the frame.
    - Note: Ethernet LAN is a broadcast network. It is possible to
      - Unicast a frame from one computer to **<u>any</u>** other computer connected to the same cable.
      - Broadcast a frame from one computer to **<u>all</u>** other computers connected to the same cable.
      - Multicast a frame from one computer to a subset of the computers connected to the same cable.

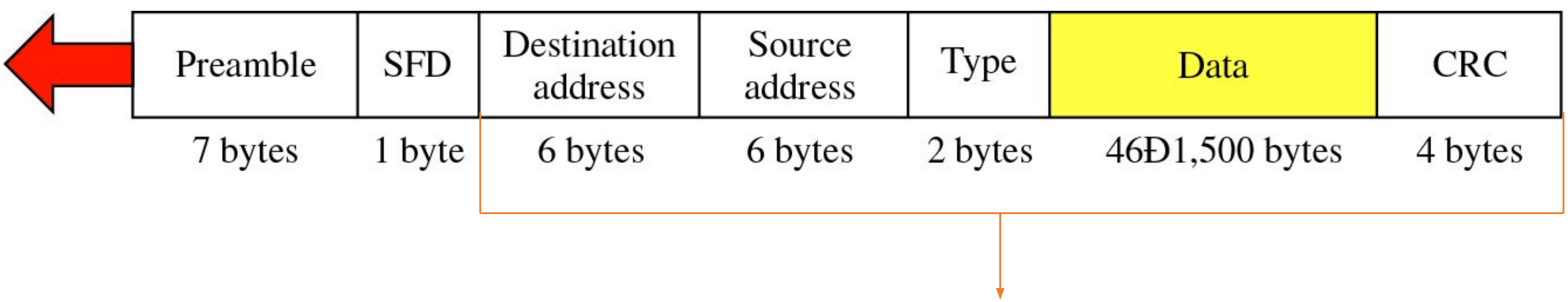
Carrier Sense Multiple Access with Collision Detection (CSMA/CD)

The basic idea:

When a station has a frame to transmit:

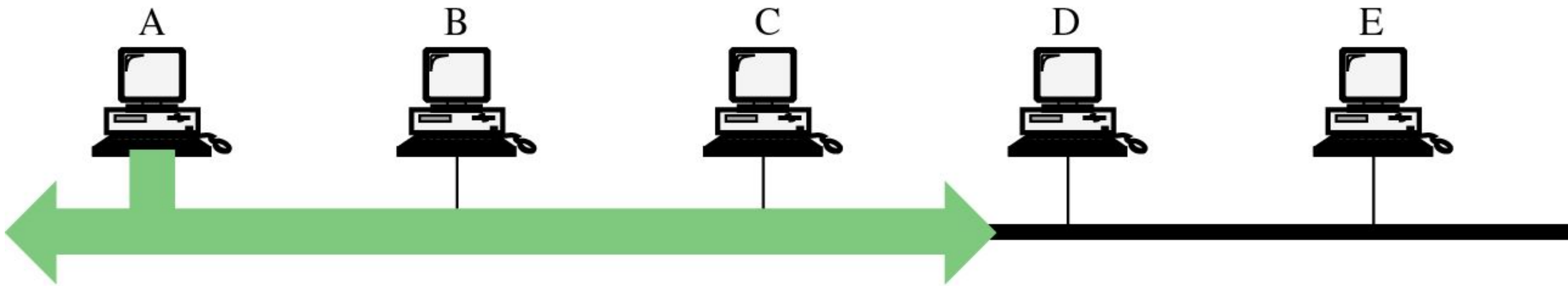
1) Listen for Data Transmission on Cable (Carrier Sense)
2) When Medium is Quiet (no other station transmitting):
   a) Transmit Frame, Listening for Collision
   b) If collision is heard, stop transmitting, wait random time, and transmit again.

Frame format

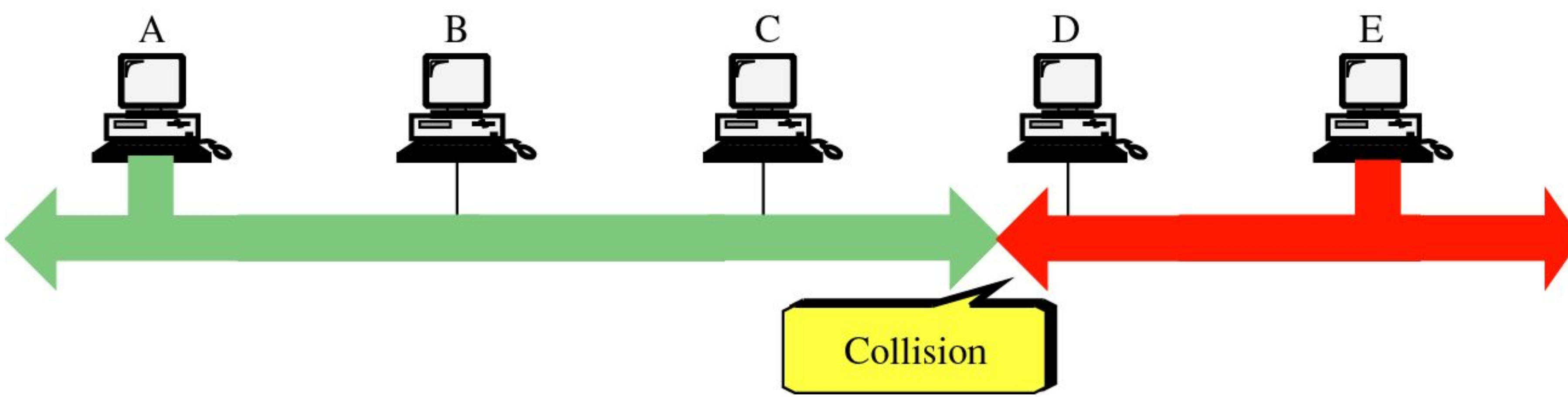


This portion must be at least 64 bytes for the Ethernet to work correctly

Computer A transmits data.



Before the signal reaches Computer E,
E transmits data. Collision occurs.



IEEE 802.3 Cabling Standards

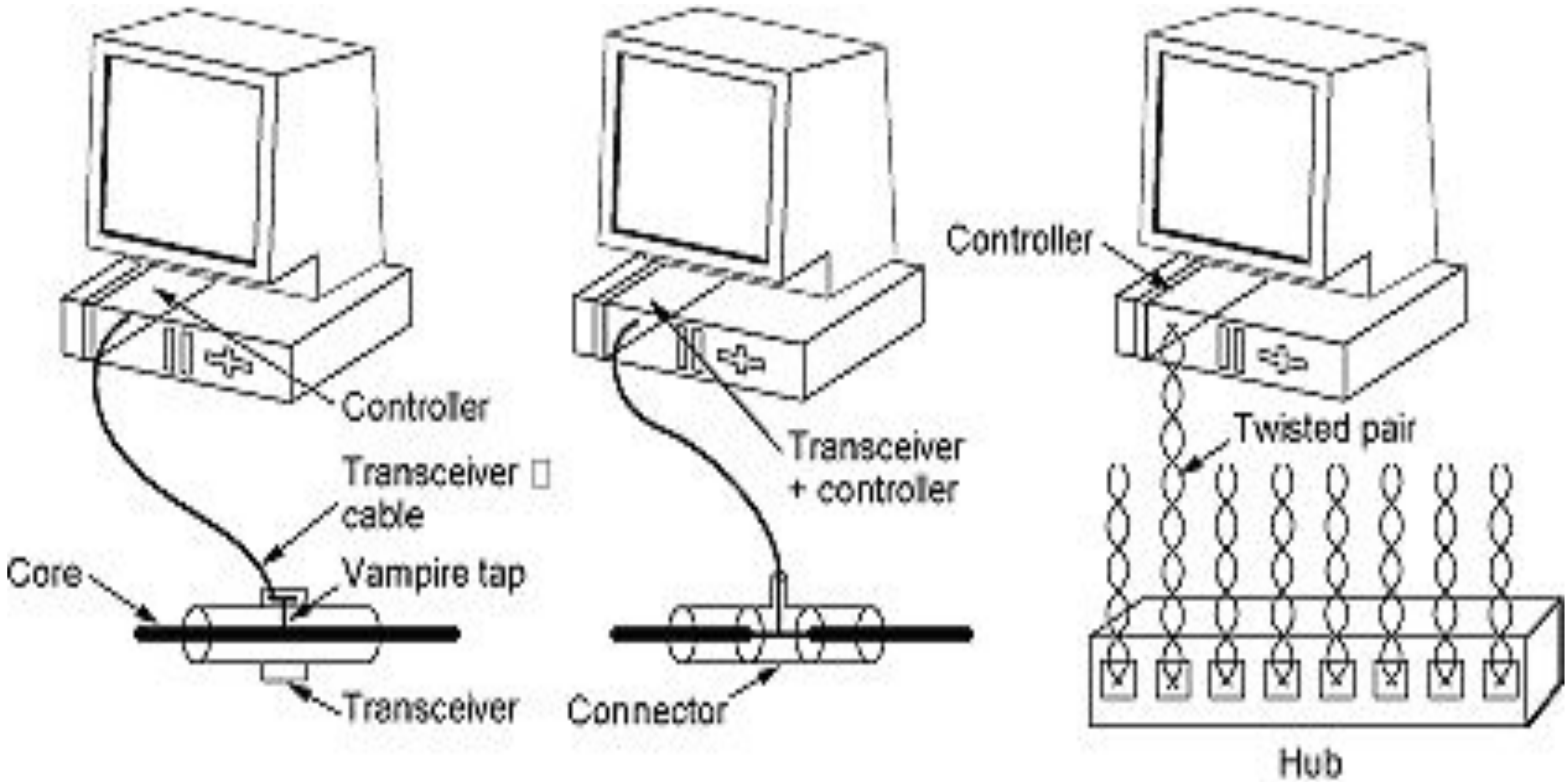
- The following is a list of the commonly used cables in 802.3 LANs.

Cable Name	Cable Type	Transmissio n Rate	Max. length before repeater needed	Max. No. of Computer
10Base5	Thick coaxial	10Mbsec	500 metres	100 per segment
10Base2	Thin coaxial cable	10Mbsec	200 metres	30 per segment
10BaseT	Twisted Pair	10/100 Mbsec	100 metres	1024 per segment
10BaseF	Fiber Optic cable	100/1000 Mbsec	2000 metres	1024 per segment

- IEEE 802.3 Cabling Standards
  - 10Base5
  - Bus topology is used.
    - Connections are made using tap to a thick coaxial cable.
    - This results in a poor connection between the computer and the coaxial cable.
  - 10Base2
    - This was the most popular 802.3 cable because it was cheap and there was a good connection between the computer and the coaxial cable.

- 10BaseT
- star topology is used.
- Stations are connected to a hub via two pairs of twisted cable.
  - Currently most popular since it is easy to maintain but is more expensive.

- 10BaseF
  - Good for high speed connections between buildings.



(a) 10Base5

(b) 10Base2

(c) 10BaseT

Types of Ethernet

1) FAST ETHERNET

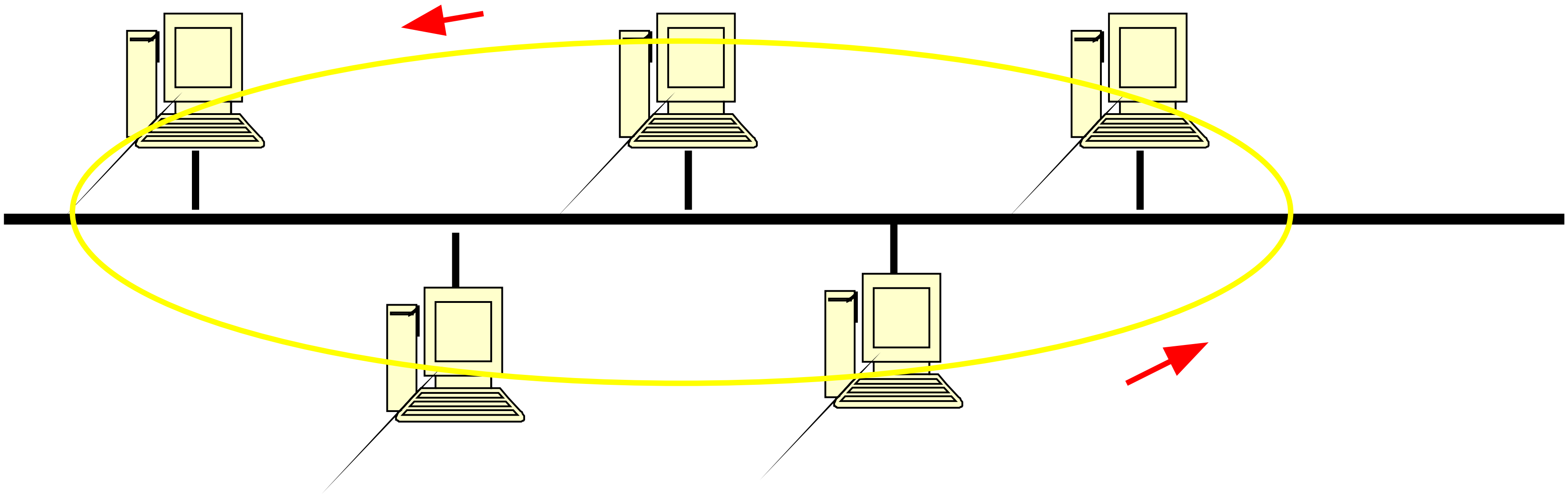
2)GIGABIT ETHERNET

-1 Gigabit

-10 Gigabit

Token Bus – IEEE 802.4

- A network which implements the modified **Token Ring** protocol over a **"virtual ring"** on a coaxial cable with a bus topology.
- It is mainly used for industrial applications.



Token-Passing Bus Access Method

- Physically, it is a Bus network. Logically, it is a Ring network
- Stations are organized as a circular doubly-linked list
- A distributed polling algorithm is used to avoid bus contention
- Token: Right of access
- Token Holder (The station receiving the token)
  - ❖ Transmit one or more MAC-frame
  - ❖ Poll other stations
  - ❖ Receive responses

Token-Passing Bus Access Method

- Non-Token Holder
  - Listen to the channel
  - Respond to a poll
  - Send Acknowledgement
- Logical Ring Maintenance
  - Ring Initialization
  - Addition to ring
  - Deletion from ring
  - Error Recovery

Normal Token Passing Operation

- In numerically descending order (60 -> 50 -> 30 -> 20 -> 60)
- Each station in the ring has knowledge of
  - TS: This Station's address (本站)
  - NS: Next Station's address (後站)
  - PS: Previous Station's address (前站)
- A station has a token holding timer to limit the time it can hold the token. This value is set at the system initialization time by the network management process.
- At the end of transmission, the token is passed to the next station.
- Once received the token, the station either starts to transmit or passes the token to the next station within one response window.

Addition of a Station

- Token holder has the responsibility of periodically (an inter-solicit-count timer) granting an opportunity for new stations to enter the logical ring before it passes the token.
- A Solicit-Successor-1 (SS1) control frame is issued with
  - DA = NS
  - SA = TS
  - Data = Null
- One response window is reserved for those stations desired to enter the logical ring and their address is between DA and SA. (If the address of the token-holder has the smallest address in the logical ring, then a Solicit-Successor-2 frame is issued)

Addition of a Station

- The station desired to enter the logical ring will respond with a Set-Successor frame with
  - DA = Token-holder address
  - SA = TS (Its address)
  - Data = TS
- The Token holder detects the event in the response window and takes appropriate actions:
  - No Response: Pass token to the next station
  - One Response: Pass token to the newly added station. The newly added station will update its NS value by recall the DA field of the previously received Solicit-Successor-1 frame.

Addition of a Station (Continued)

- Multiple Responses:
  - A Resolve-Contention frame is issued by the token-holder with
    - DA = XX, SA = TS, Data = Null
  - A station desired to enter the logical ring will response with a Set-Successor frame as before at the K-th window, where K is determined by the value of the first two bits of its address. However, if the channel is detected busy before the K-th window, it will give up.
  - If no valid Set-Successor frame is received by the token-holder, the token-holder will issue another Resolve-Contention frame.
  - Now only those stations involved in the contention may try again. The value of K now is determined by the next two bits.
  - The above procedure is repeated until a valid Set-Successor frame is received by the token-holder. A new station is thus successfully added to the logical ring.

Deletion of a Station

- The station wishes to be deleted may wait until it receives the token, then sends a Set-Successor frame to its predecessor, with
- DA = PS
  - SA = TS
  - Data = NS
- The previous station once receives the Set-Successor frame will modify its NS and send a token to its new next station.
- The next station once receives the Set-Successor frame will modify its PS accordingly.
- After these two modifications, the station is removed from the logical ring automatically.
- If the station fails, it will not receive the token. This will be detected by the token-sender as explained later.

Fault Management

- One of the most important issues of the token-bus protocol is to maintain the logical ring under the following possible conditions:
  - Multiple Tokens
  - Unaccepted Token
  - Failed Station
  - Failed Receiver
  - No Token

Multiple Tokens

- Cause:
  - Noise
  - Duplicate Address, each one may "receive" a token
- Detection:
  - While holding the token, the station may hear a frame on the bus which indicating that another station also has a token.
- Action:
  - Drop the token
  - If all stations drop the token, the network becomes the case of no token (see the procedure of handling no token later)

Unaccepted Token or Failed Station

- Cause
  - The token passed to the next station may be garbled
  - The next station fails
- Detection:
  - No response (Channel is idle) in one response window
- Action:
  - Try to pass token one more time
  - It still no response, then the next station is assumed to have failed
  - The token holder then issues a Who-Follows frame with
    - DA = XX, SA = TS, Data = NS

Unaccepted Token or Failed Station

- All other stations once received Who-Follows frame will compare the data with its PS value. If there is a match, it will issue a Set-Successor frame back. Three response windows are reserved after Who-Follows. The first two are needed to make a comparison.
- If no response to the Who-Follows frame, the above procedure will be tried one more time.
- If still no response to the Who-Follows frame, then it could be that the next station to the next station has also failed.
- The token-holder will try to establish the ring by issuing a Solicit-Successor-2 frame, with
  - DA = TS, SA = TS, Data = Null.

Unaccepted Token and Failed Station

- DA=SA=TS implies that every station is invited to respond. Two response windows are reserved after this frame.
- The first response window is reserved for stations whose address is less than the sender.
- The second response window is reserved for stations whose address is greater than the sender.
- The procedure of add a station is then used.
- If still no response to the Solicit frame, then either all stations have failed (Left the ring) or its own receiver has failed (so it cannot listen).
- If the only one station has something to send, it sends the data. Then repeat the token passing process. Otherwise, listen to the channel.

No Token or Initialization

- Cause:
  - The Token-holder station fails
  - The token is destroyed
  - Network Initialization
- Detection:
  - No channel activity has been heard for a certain amount of time (Bus-Idle Timer expired)
- Action:
  - Any station when its Bus-Idle timer is expired will issue a Claim-Token frame, with
    - DA = XX, SA = TS, Data = Any value with (0,2,4,6) slot times depending on its address

No Token or Initialization

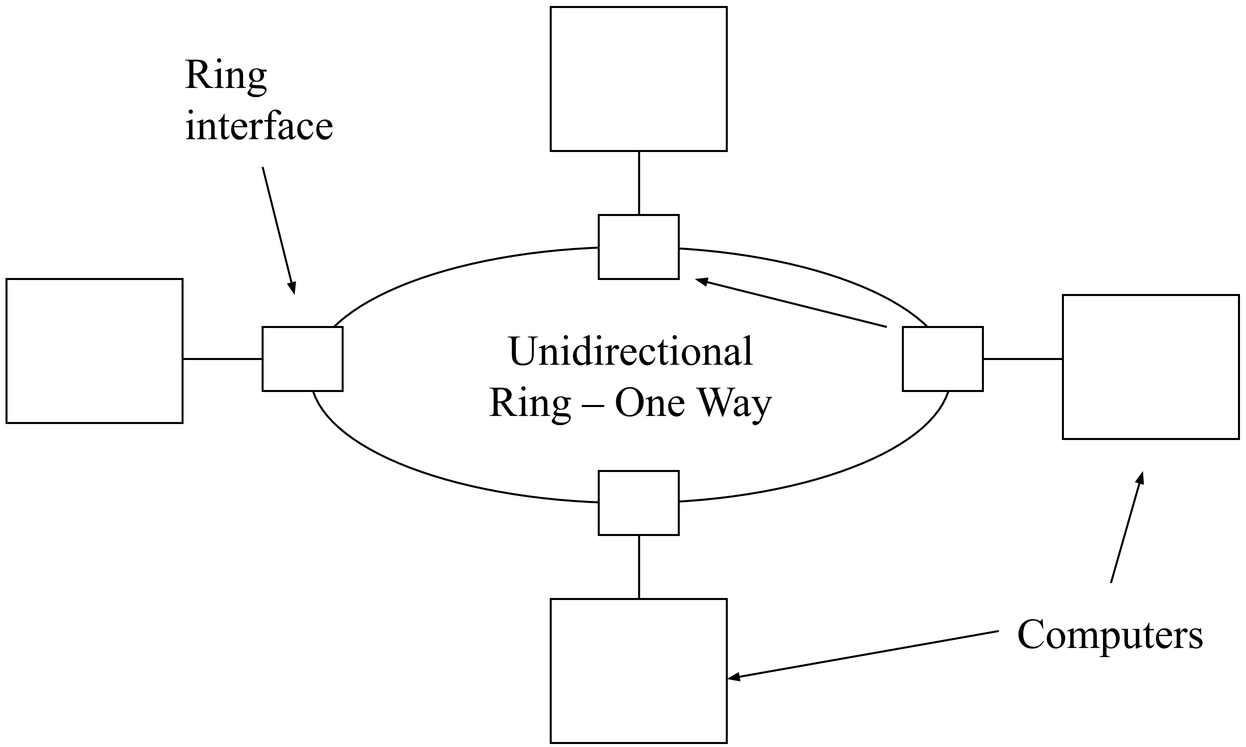
- The station with the greatest address will get the token. This is done by comparing the address. Two bits of the address are compared at a time.
- In each pass, only those stations who transmitted the longest frame on the previous pass try again.
- The station that succeeds on the last pass considers itself the token holder.
- The difference is 2 slots in the frame padding. The station waits one slot for its or other frame to pass. It then samples the channel at the second slot.
- The logical ring can then be established by issuing Solicit-Successor frames as described before.

Priority Mechanism

- In the control field of data frame, three bits are reserved to indicate frame priority.
- Only four access classes are considered
  - 6 : The highest priority
  - 4
  - 2
  - 0 : The lowest priority
- Hi-Pri-Token-Hold-Time: To avoid one station dominating the network, an upper bound is set in each station to determine the maximum time that the highest priority frame can hold the token.
- Each lower access class in the station has a Target token Rotation Time (TRT).

802.5 Token Ring LANs

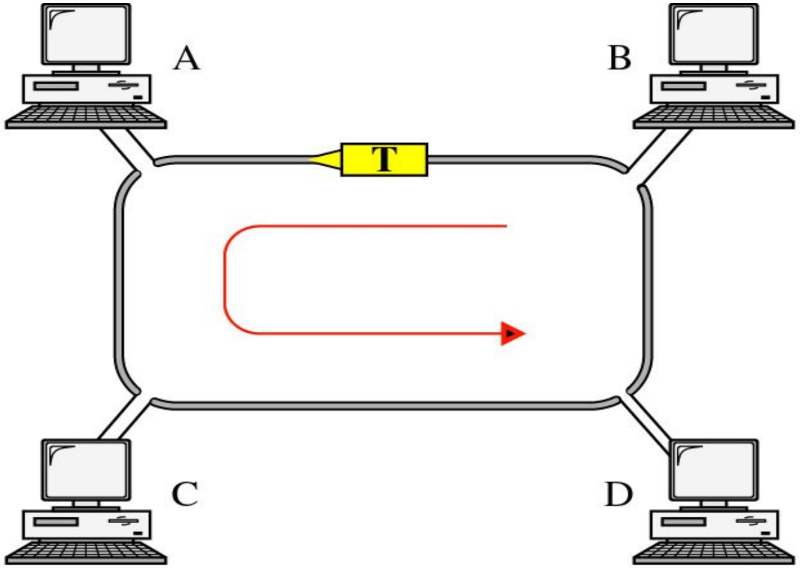
– A Token Ring LAN consists of a collection of ring interfaces connected by point-to-point lines.



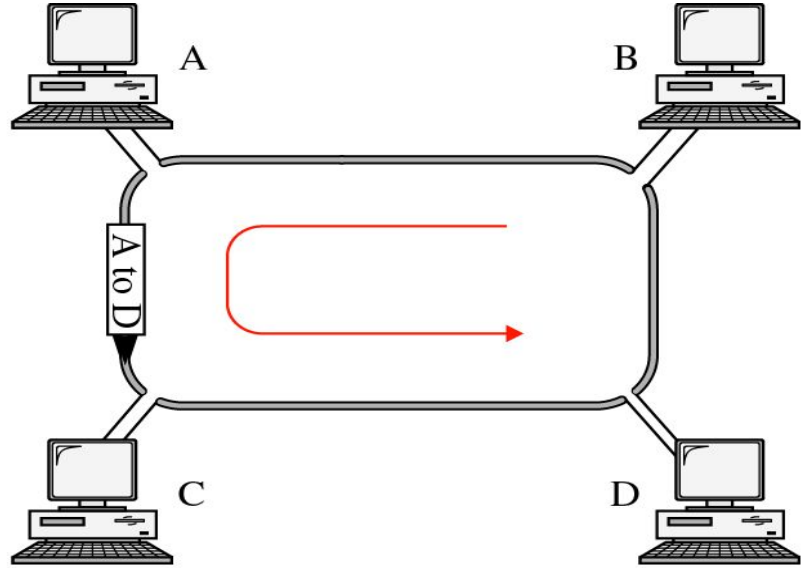
- IEEE 802.5 Token Ring LANs
  - The MAC sub layer uses Token Ring Technology.
  - In a Token Ring LAN, a special bit pattern called the token circulates around the ring whenever all computers are idle.

When a computer wants to transmit:

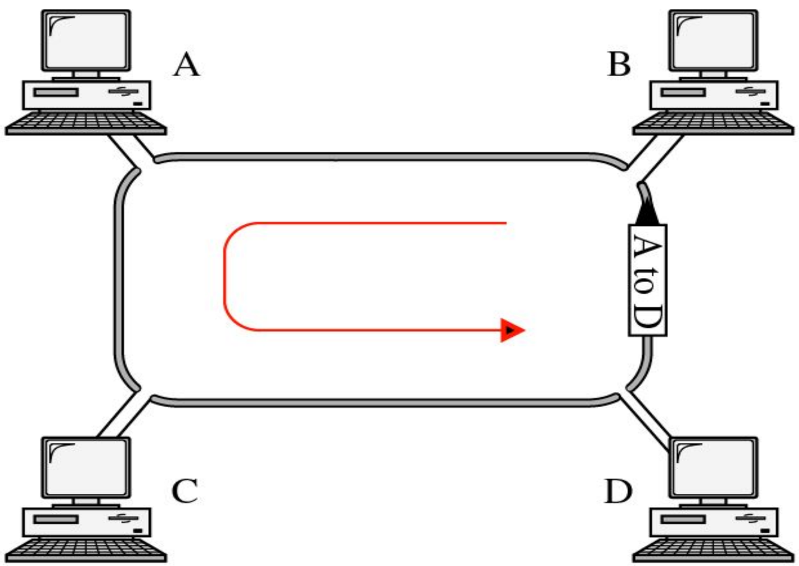
- It waits for the token to arrive.
- When it arrives, it removes the token from the ring. There is only one token so only one computer can transmit at any one time.
- The computer can now transmit its frame on its output link.
- This frame will now propagate around the ring until it arrives back at the sender who removes the frame from the ring.
- The sender then regenerates the token and passes it to the next computer (restarting the above steps).



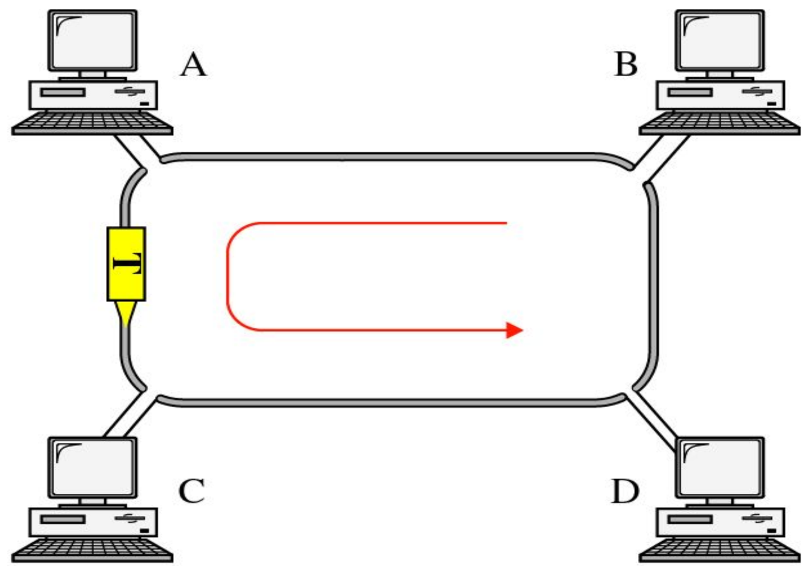
a. Token is traveling along the ring.



b. Station **A** captures the token and sends its data to D.



c. Station D copies the frame and sends the data back to the ring.



d. Station **A** receives the frame and releases the token.

IEEE 802.5 Frame Format

Bytes	3	2 or 6	2 or 6		4	1	1
	Start of Frame Delimiter	Destination Address	Source Address	Data – No Limit	Checksum Same as 802.3	End Delimiter	Frame Status

Token 3 Bytes

Difference between Start of Frame and Token is only 1 bit in 3rd byte.

- Start of Frame and End Delimiters mark the beginning and ending of a frame.
- Destination Address, Source Address and Checksum are used in a similar fashion to IEEE 802.3 Ethernet.

- Comparison of 802.3 ‘Ethernet’ and 802.5 ‘Token Ring’
  - Generally they have similar technology with similar performance.
  - 802.3 Ethernet – Advantages
    - Widely used at present. People are experienced in using this technology.
    - Simple Protocol. New computers can be added with having to bring the network down.
    - Almost zero delay at low load, there is no need to wait for a token, you can transmit when ready.

- 802.3 Ethernet – Disadvantages
  - The electronics is more complicated for carrier sense and collision detection.
  - The smallest frame must hold 64 bytes, this means there is a substantial overhead if you are only transmitting a single character from your machine.
  - Ethernet is non-deterministic system (possibility of repeated collisions). This means that Ethernet is not suitable for network applications that require guaranteed delivery times.
  - Poor performance at high loads as there can be lots of collisions reducing the number of messages that are successfully transmitted.

- Comparison..........
  - 802.5 Token Ring – Advantages
    - Token Ring uses point-to-point connections between ring interfaces so that the electronic hardware can be fully digital and simple. There is no need for collision detection.
    - Can use any medium twisted pair is cheap and easy to install but could equally use fiber optic if available.
    - Throughput excellent at high loads since there is no possibility of collisions unlike 802.3.

- ## 802.5 Token Ring – Disadvantages
  - Computers must wait for the token to arrive, therefore at load, a computer is delayed before sending.
  - Each token ring has a monitor computer, to look after the ring. If the monitor computer failed, the remaining computers would have to wait until it is replaced before being able to continue.

What is 802.11 ??

- 802.11 refers to a family of specifications developed by the IEEE for wireless LAN technology. 802.11 specifies an over-the-air interface between a wireless client and a base station or between two wireless clients.
- The IEEE accepted the specification in 1997.

IEEE 802.11…….

- Defines two services:

1- the basic services set (BSS)

2- the extended service set (ESS)

BSS……

- Made of mobile wireless stations and an optional central base station, known as Access point (AP).
- Without AP BSS can’t send data to other BSS s.
- With AP is called an “infrastructure network”.

ESS…….

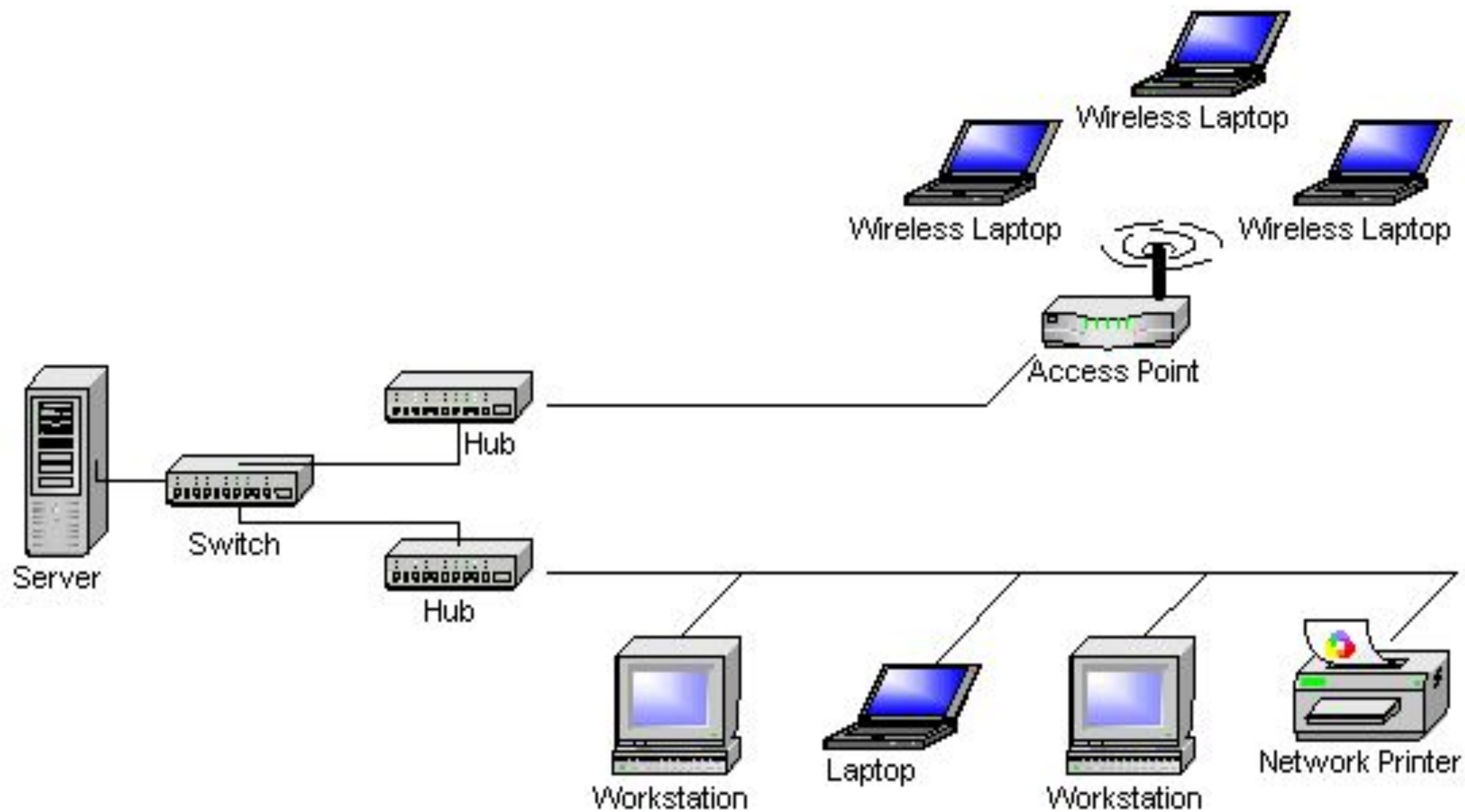
-Made up of two or more BSSs with APs.

-BSS s are connected through distribution system, which is usually a wired LAN.

-uses two types of stations: mobile and stationary.

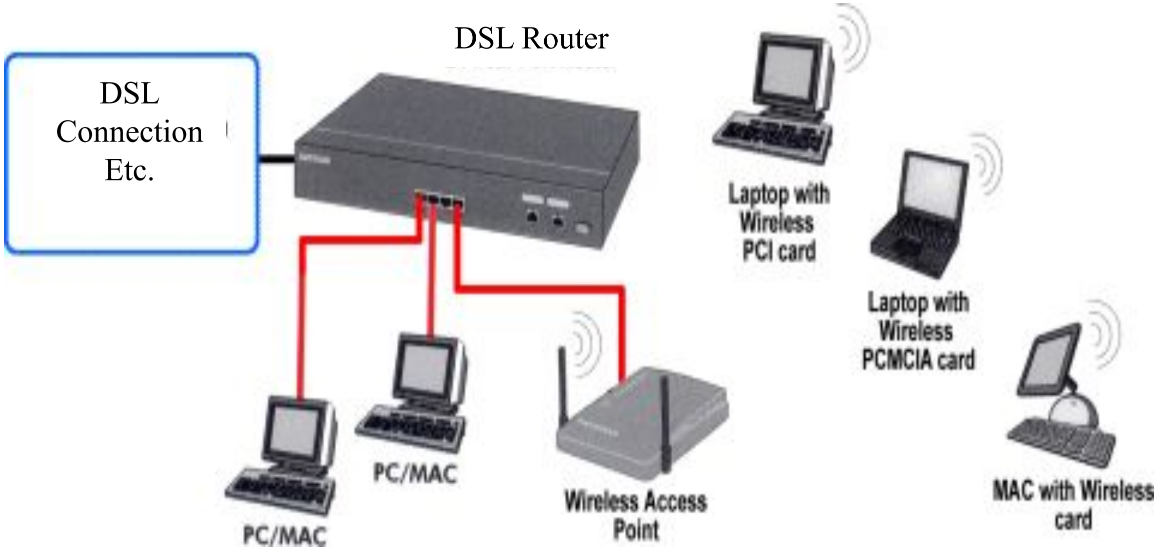
Wireless LAN Topology

- Wireless LAN is typically deployed as an extension of an existing wired network as shown below.



Wireless LAN Topology

- Here is an example of small business usage of Wi-Fi Network.



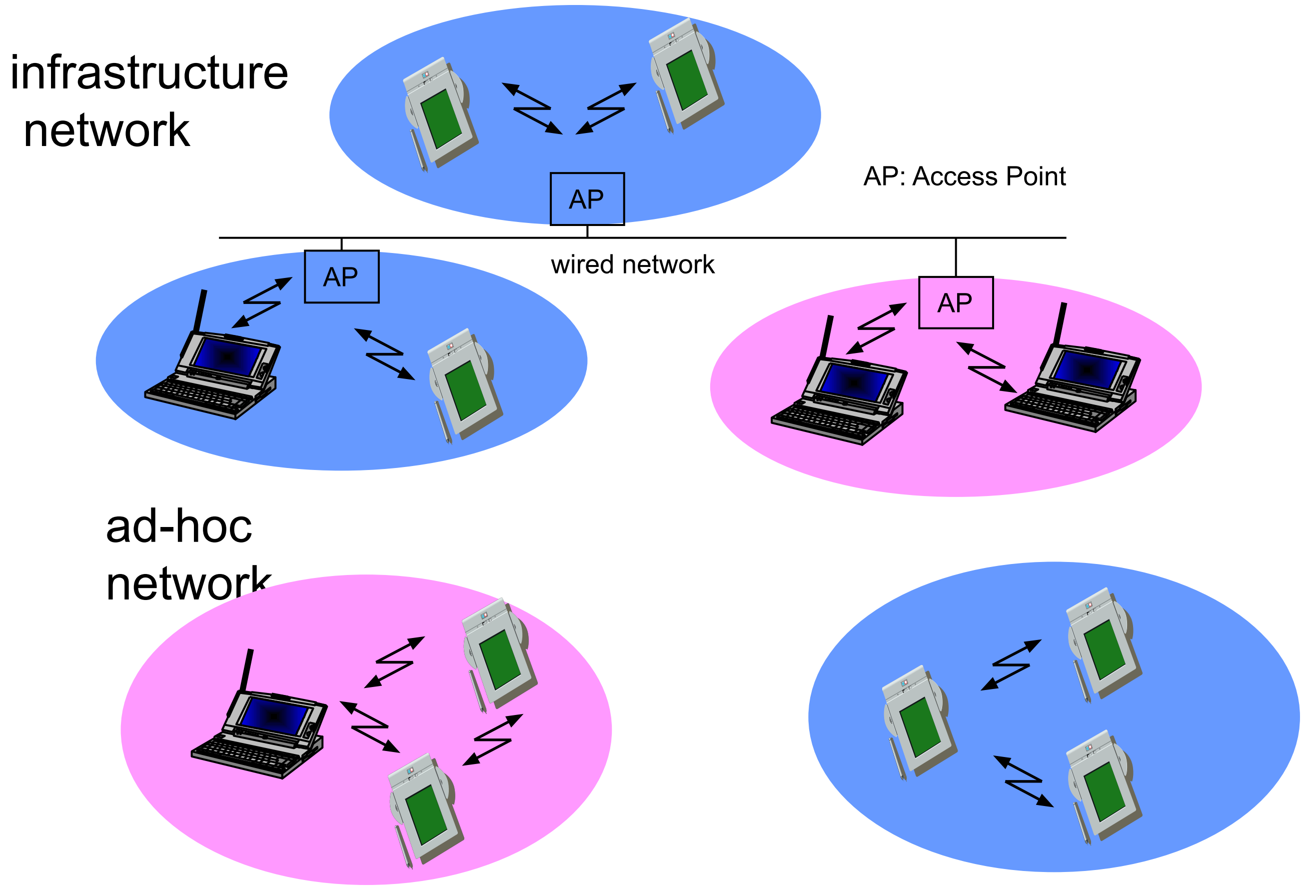
The DSL router and Wi-Fi AP are often combined into a single unit

802.11 Wireless LAN Working Group(1/2)

- Types
  - Infrastructure based
  - Ad-hoc

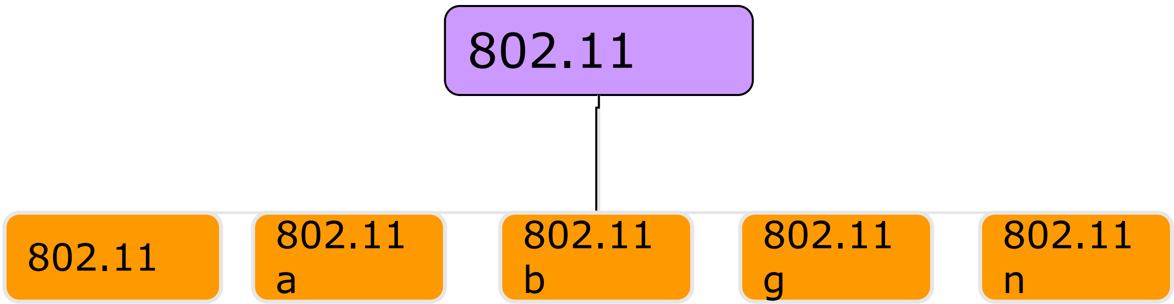
- Advantages
  - Flexible deployment
  - Minimal wiring difficulties
  - More robust against disasters (earthquake etc)

- Disadvantages
  - Low bandwidth compared to wired networks (1-10 Mbit/s)
  - Need to follow wireless spectrum regulations
  - Not support mobility


infrastructure network
AP
AP: Access Point
wired network
AP
AP
ad-hoc network

802.11 Wireless LAN Working Group(2/2)

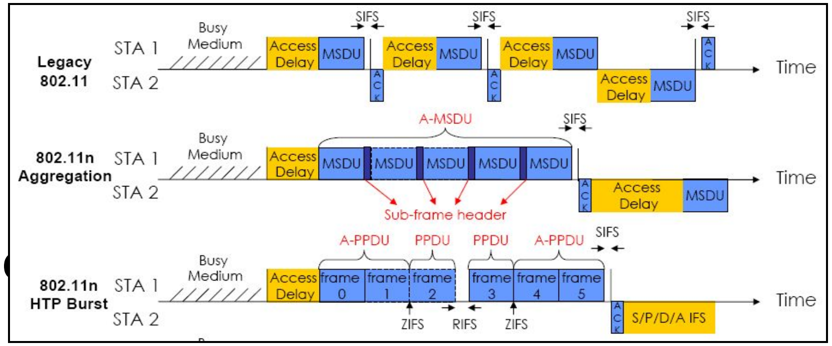
- Working Groups summary



Protocol	Release date	Op. Frequency	Data rate (Max)	Range (indoor)	Range (outdoor)
Legacy	1997	2.5~2.5 GHz	2 Mbit/s		
802.11a	1999	5.15~5.35/5.47~5.725/5.725~5.875 GHz	54 Mbit/s	~25 m	~75 m
802.11b	1999	2.4~2.5GHz	11 Mbit/s	~35 m	~100 m
802.11g	2003	2.4~2.5GHz	54 Mbit/s	~25 m	~75 m
802.11n	2007	2.4GHz or 5GHz	540 Mbit/s	~50 m	~125 m

802.11n Working Group

- What is the 802.11n?
  - Uses MIMO radio technology and OFDM as a basis
  - Anywhere from 100Mbps to 600Mbps depending on implementation
  - Support both 2.4 GHz and 5 GHz
  - Use muiltiple stream

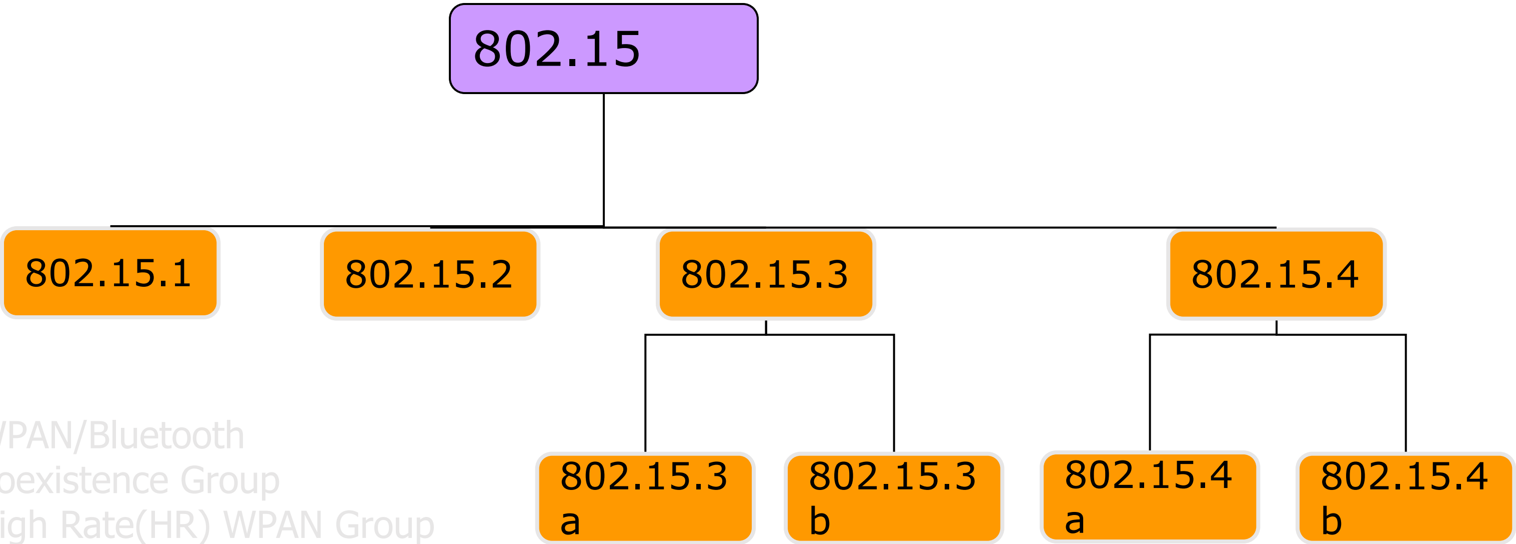


- 802.11n increase transmission efficiency
  - Cutting guard band time in half
  - Reducing the number of pilot carrier, for data
  - Aggregating frames and bursting
  - Using a 40MHz instead of a 20MHz channel

30~50% => 70%

802.15 Wireless Personal Area Network(WPAN) Working Group

❑ Working Groups summary



- 802.15.1 : WPAN/Bluetooth
- 802.15.2 : Coexistence Group
- 802.15.3 : High Rate(HR) WPAN Group
  - 802.15.3a : WPAN HR Alternative PHY Task Group
  - 802.15.3b : MAC Amendment Task Group
- 802.15.4 : Low Rate(LW) WPAN Group(Zigbee)
  - 802.15.4a : WPAN Low Rate Alternative PHY
  - 802.15.4b : Revisions and Enhancements
- UWB Forum

Bluetooth

- What is the Bluetooth?
  - Radio modules operate in 2.45GHz. RF channels:2420+k MHz
  - Devices within 10m of each other can share up to 1Mbps
  - Projected cost for a Bluetooth chip is ~$5.
  - Its low power consumption
  - Can operate on both circuit and packet switching modes
  - Providing both synchronous and asynchronous data services

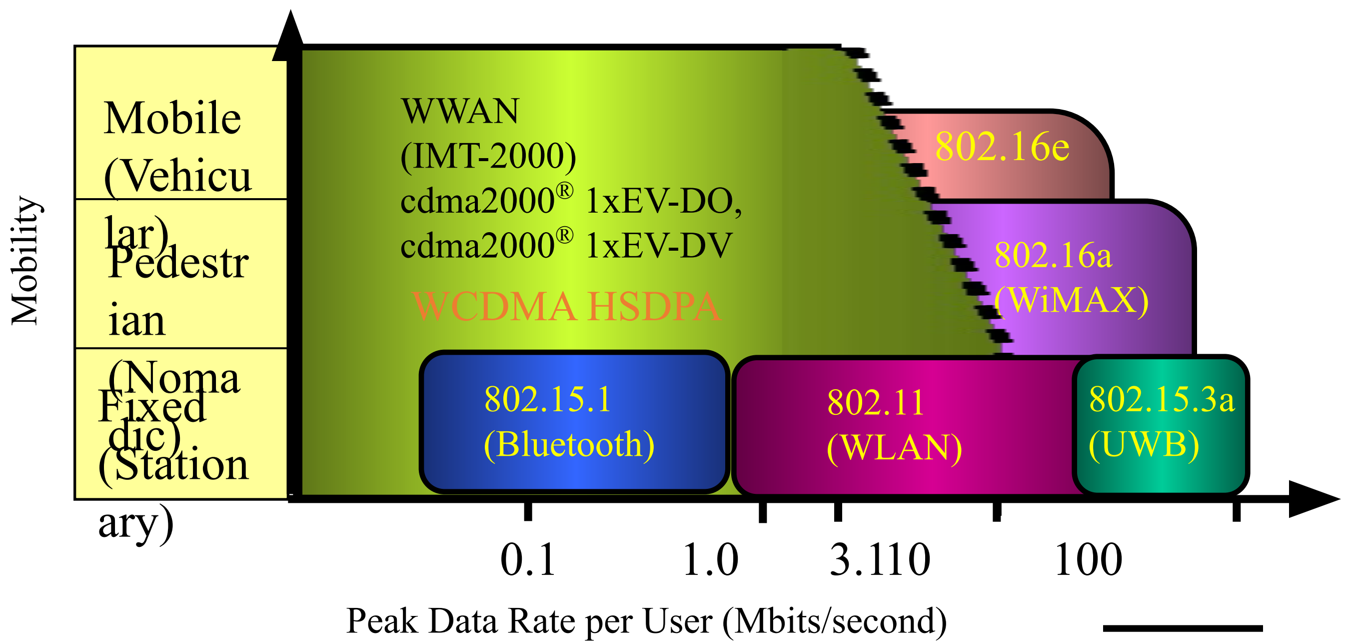
	Bluetooth	IEEE 802.11A	UWB
frequency	2.4Ghz	5GHz	3.1~10.6GHz
MAX data rate	1Mbps	54Mbps	100Mbps~1Gbps
Range	5~10m	35~50m	10~30m
The number of channel	79	12	…..

Bluetooth versions

- **Bluetooth 1.0 and 1.0B**
  - Versions 1.0 and 1.0B had many problems
    - Manufacturers had difficulty making their products interoperable.
- **Bluetooth 1.1**
  - Many errors found in the 1.0B specifications were fixed.
  - Added support for non-encrypted channels.
  - Received Signal Strength Indicator (RSSI).
- **Bluetooth 1.2**
  - Faster Connection and Discovery
  - Use the *Adaptive frequency-hopping spread spectrum (AFH)*
    - improves resistance to radio frequency interference
  - Higher transmission speeds in practice, up to 721 kbps
- **Bluetooth 2.0**
  - This version, specified November 2004
  - The main enhancement is the introduction of an enhanced data rate (EDR) of 3.0 Mbps.
  - Lower power consumption through a reduced duty cycle.
  - Simplification of multi-link scenarios due to more available bandwidth.
- **Bluetooth 2.1**
  - A draft version of the Bluetooth Core Specification Version 2.1 + EDR is now available

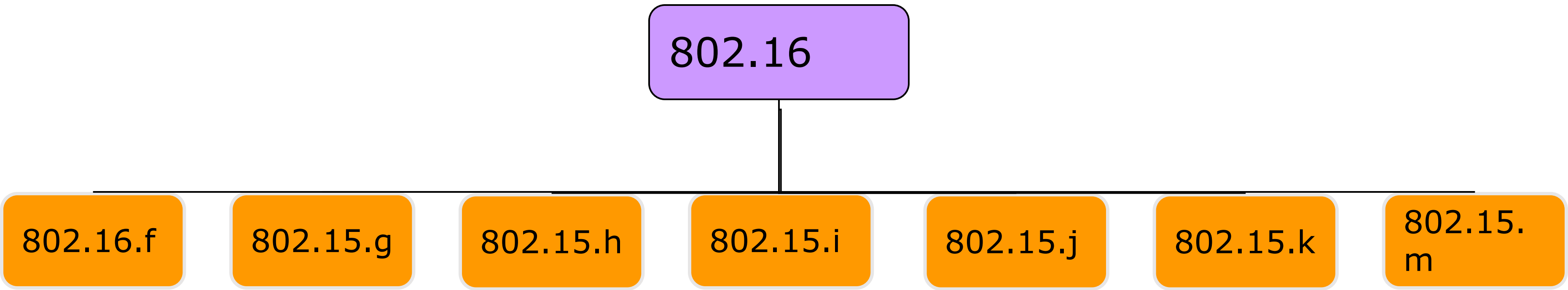
802.16 Broadband Wireless Access(BWA) Working Group(1/2)

- IEEE 802.16
  - Be was established by IEEE Standards Board in 1999, aims to prepare formal specifications for the global deployment of broadband Wireless Metropolitan Area Network.
  - A unit of the IEEE 802 LAN/MAN Standards Committee.
  - A related technology Mobile Broadband Wireless Access(MBWA)



802.16 Broadband Wireless Access(BWA) Working Group(2/2)

- Working Groups summary



- 802.16f : Management Information Base
- 802.16g : Management Plane Procedures and Services
- 802.16h : Improved Coexistence Mechanisms for License-Exempt Operation
- 802.16i : Mobile Management Information Base
- 802.16j : Multihop Relay Specification
- 802.16k : Bridging of 802.16
- 802.16m : Advanced Air Interface. Data rates of 100 Mbps for mobile applications and 1 Gbps for fixed applications.

802.11 Family Members

- There are several specifications in the 802.11 family:

- **802.11**

  - Applies to wireless LANs and provides 1 or 2 Mbps transmission in the 2.4 GHz band using either frequency hopping spread spectrum (FHSS) or direct sequence spread spectrum (DSSS).

- **802.11a**

  - An extension to 802.11 that applies to wireless LANs and provides up to 54 Mbps in the 5GHz band. 802.11a uses an orthogonal frequency division multiplexing encoding scheme rather than FHSS or DSSS.

- **802.11b**
  - (also referred to as 802.11 High Rate or Wi-Fi) is an extension to 802.11 that applies to wireless LANs and provides 11 Mbps transmission (with a fallback to 5.5, 2 and 1 Mbps) in the 2.4 GHz band. 802.11b uses only DSSS. 802.11b was a 1999 ratification to the original 802.11 standard, allowing wireless functionality comparable to Ethernet.
- **802.11g**
  - Applies to wireless LANs and provides 20+ Mbps in the 2.4 GHz band.

Why Choose? A vs B vs G

Wireless Technology Comparison Chart

Wireless Standard	802.11b	802.11a	802.11g
Popularity	Widely adopted. Readily available everywhere.	New technology.	New technology with rapid growth expected.
Speed	[11 Mbps] Up to 11Mbps (note: cable modem service typically averages no more than 4 to 5Mbps).	[54 Mbps] Up to 54Mbps (5X greater than 802.11b).	[54 Mbps] Up to 54Mbps (5X greater than 802.11b).
Relative Cost	Inexpensive.	Relatively more expensive.	Relatively inexpensive.
Frequency	[2.4 GHz] More crowded 2.4GHz band. Some conflict may occur with other 2.4GHz devices like cordless phones, microwave ovens, etc.	[5 GHz] Uncrowded 5GHz band can coexist with 2.4 GHz networks without interference.	[2.4 GHz] More crowded 2.4GHz band. Some conflict may occur with other 2.4GHz devices like cordless phones, microwave ovens, etc.
Range	[100-150] Good Range. Typically up to 100-150 feet indoors, depending on construction, building materials, room layout.	[25-75] Shorter range than 802.11b & 802.11g. Typically 25 to 75 feet indoors.	[100-150] Good Range. Typically up to 100-150 feet indoors, depending on construction, building materials, room layout.
Public Access	The number of public "hotspots" is growing rapidly, allowing wireless connectivity in many airports, hotels, college campuses, public areas, and restaurants.	None at this time.	Compatible with current 802.11b hotspots (at 11Mbps). Also, it is expected that most 802.11b hotspots will quickly convert to 802.11g.
Compatibility	[OK 802.11b] Widest adoption.	[OK 802.11a] Incompatible with 802.11b or 802.11g.	[OK 802.11b 802.11g] Interoperates with 802.11b networks (at 11Mpbs). Incompatible with 802.11a.

Benefits of A vs B vs G

802.11b

Wireless-B

- Lowest price
- Excellent signal range
- Coverage penetrates most walls
- Works with public hotspots

802.11a

Wireless-A

- Supports more users per room
- Unaffected by interference from 2.4GHz devices
- Can co-exist with B and G networks
- Coverage limited To one room

802.11g

Wireless-G

- Best value - only 10% premium for 5 times the speed of Wireless-B
- Compatible with Wireless-B networks and hotspots
- Excellent signal range
- Coverage penetrates most walls